

# Xinlin Zhang

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## Career Objective

Software | Data | AI Engineer

Seasoned software engineer with a First-Class Honored Master's Degree of Computer and Information Science in AUT. Passionate about real-world software applications. Seeking to join a team in which I can apply my expertise in building innovative and cutting-the-edge systems.

## Skills Summary

- **Programming:** Proficient in Python ,SQL, familiar with PyTorch
- **Strong Computer Science Background:** expertise on Computer Science, OS, Network, Algorithm and Database.
- **Seasoned Software Engineer:** many years' software developing experience in C/C++ and Python, familiar with Linux and Shell.
- **AI:** Experienced in designing, implementing, training and evaluating various deep learning models from scratch by using PyTorch, specialized in building customized CNN architectures. Familiar with the R-CNN and YOLO series models for object detection. Hands-on experience in deploying self-hosted LLMs and deep understanding of prompt engineering.
- **[Personal Projects Portfolio](https://github.com/xinlin-z/portfolio)** (github.com/xinlin-z/portfolio)

## Professional Experience

### **AI Software Engineer (Part Time Research Support), TOMRA Food (New Zealand), 09/2026 – Present**

- Work in Mint project for internal fruit grading

### **AI Software Research Engineer (Intern), TOMRA Food (New Zealand), 11/2025 – 08/2026**

- Working in the Data Scientist Team as an Intern.
- Design, Implement, Train and Evaluate highly modernized, customized and configurable deep learning models for multi-class image segmentation tasks. These models have fixed input and output, and need to be run in a memory-constrained real-time environment. The model's performance is good enough to test on the customer's site. (Python, PyTorch)
- Design and implement wrapper model to include post-processing computations and realized end-to-end pipeline. (PyTorch)
- Automate Label Studio into working flow, realize interactive model-assistant data labelling, and build training and validation dataset. (Python, Shell, Label Studio)
- Implement the whole working codebase, including models, data augmentation algorithms, training and fine-tuning scripts, and all other tools. (Python, PyTorch, Shell, Javascript, FastAPI)

### **Senior System Software Engineer, Astute-Tec (China), 2021 – 2023**

- Designed and implemented a highly optimized and configurable BGRA image comparison algorithm and a hybrid H.264/JPEG transportation protocol, reducing CPU utilization from over 20% to below 5% in a cloud desktop system during static or minimal screen activity. (C++, Python)
- Designed and implemented a priority-based multi-TCP schedule algorithm, significantly improving user responsiveness and fluidity in a highly loaded cloud virtual desktop environment. (C++)
- Recipient of the “Network and Video Codec Expert Award”, May 2022.

### **Software Engineer, VP, Co-founder, GWorld (China), 2014 – 2021**

- Solely designed and implemented an auto-production system for manufacturing company proprietary optical modules ( $\leq 10G$ ) and OEM modules. (Python, SQLite)

### **Software Engineer, ZTE R&D Center (China), 2006 – 2013**

- Developed and maintained many features in the TCP/IP stack (such as Ethernet, VLAN, Routing Protocol, ACL, and RTP) which were integrated into many different telecommunication products. (C)
- Recipient of the Department's “Best Programmer Award”, December 2009.

## **Education**

### **Auckland University of Technology (New Zealand), Feb. 2025 – Jul. 2026**

Master of Computer and Information Science. Graduated with Honours (First Class). Straight-A.

#### **Coursera.org:**

AI Agent Developer (Vanderbilt University), [Certificate](#) earned in Jun. 2025

Mathematics for Machine Learning and Data Science (DeepLearning.AI), [Certificate](#) earned in Feb. 2025

### **Northeastern University (China), Sep. 1998 – Jul. 2002**

Bachelor of Engineering Degree in Management Information Systems (under the department of Computer Science)

## **Interests**

Reading, Coding, Hiking and Green Tea.

## **Reference**

Patrick Lindegger

Data Scientist in TOMRA Food

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Yanbin Liu

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